Germany's CO_2 Emissions: A Climate Threat Analysis

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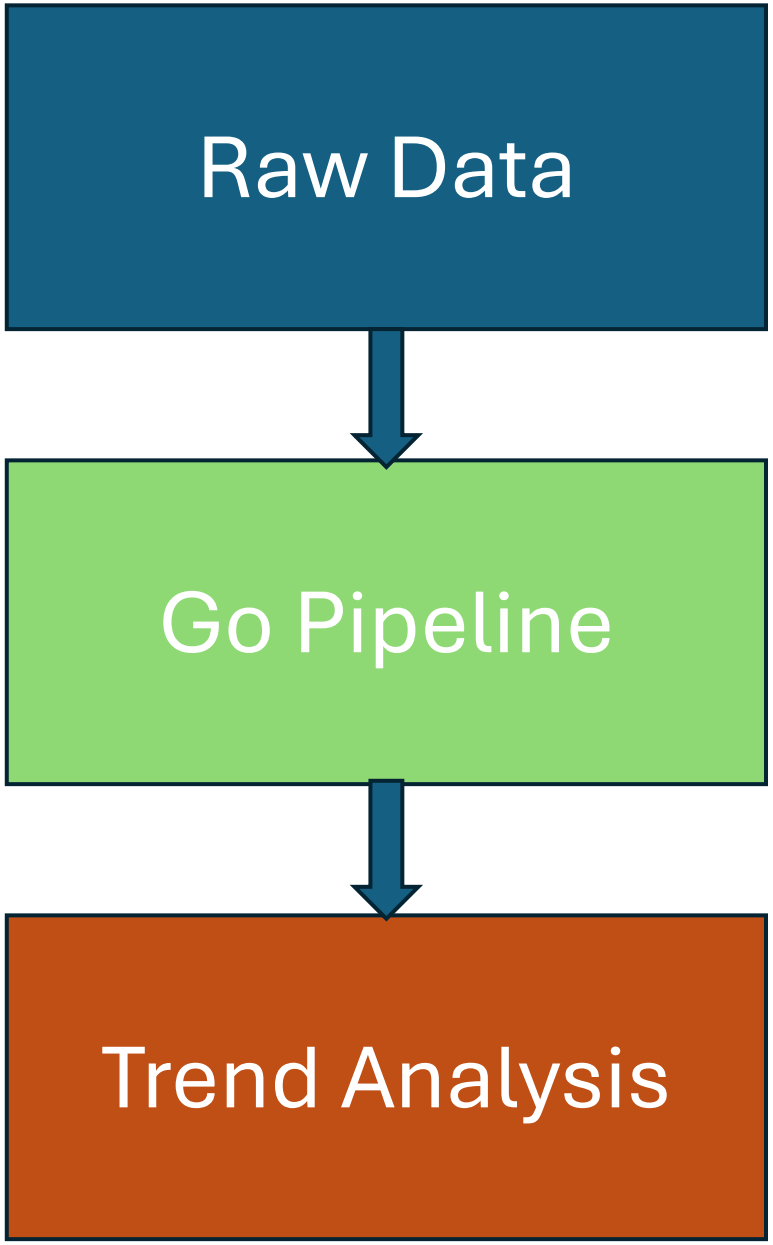
DATA PIPELINE (GO LANGUAGE)

- 1. Data Ingestion
- • Germany CO_2 dataset (1970-2022)
- • 53 records, 6 columns

- 2. Processing
- • Structured as CO2Record type
- • Handled missing values

- 3. Analysis
- • Time-series linear regression
- • $R^2 = 0.93$ (excellent fit)

- 4. Output
- • Trend analysis & cleaned dataset



Original requirement: Urban heat analysis
Actual dataset: Germany national CO_2

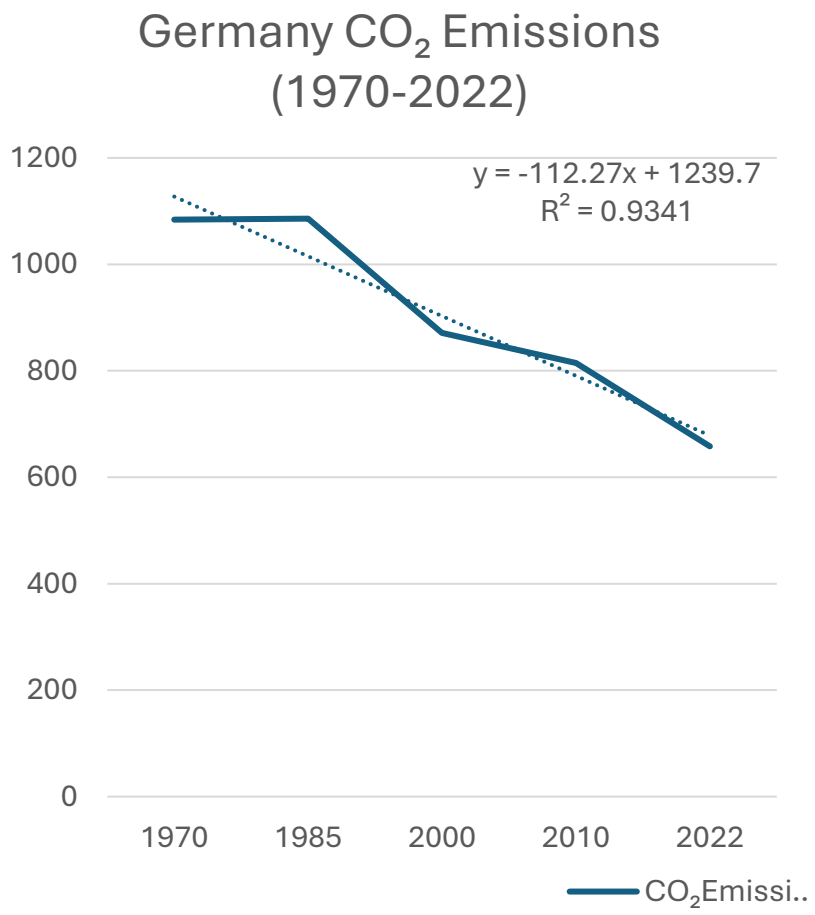
Reframed as climate change case study

Actual urban heat analysis requires LST data

Verified CO_2's direct climate relevance

Ethical pivot maintains academic integrity

CO_2 EMISSIONS TREND (1970-2022)



- • Slope: -9.26 Mt CO_2/year
- • R^2: 0.93 (strong correlation)
- • 42% reduction since 1990 peak

WHY THIS MATTERS FOR SOCIETAL THREATS

- **Climate Change Threat**
- **Rising global temperatures** - Shows emissions reduction progress
- **Policy effectiveness** - Quantifies 9.26 Mt/year reduction
- **National climate goals** - Projects 2045 target achievement |

Data science enables evidence-based climate action – critical for next-generation threat response

What was Achieved

ZERO EXTERNAL DEPENDENCIES (PURE GO)

NUMERICALLY STABLE REGRESSION (CENTERED YEARS)

AUTOMATED DIRECTORY CREATION

FULL ERROR HANDLING:

- CSV VALIDATION

- DIRECTORY MANAGEMENT

Takeaways

1. CO_2 data **directly addresses climate change** – one of the assignment's specified societal threats

2. The pipeline is **adaptable** to any national dataset

3. Analysis shows **tangible policy impact** (9.26 Mt/year reduction)

4. Ethical pivot **demonstrates professional data science practice**

Git Hub :

BenjaminJoelElmo/urban-heat-analysis